CLASS 10 — MATHS · FORMULA SHEET

Chapter 9: **Some Applications of Trigonometry** | Quick revision, one page

Terms

- **Line of sight**: line from the eye to the object. **Angle of elevation**: angle between the line of sight and the horizontal when the object is above the horizontal.
- **Angle of depression**: angle between the line of sight and the horizontal when the object is below the horizontal. Angle of depression from the top = angle of elevation from the object (alternate angles).

Ratios used

- $\sin\theta = \dfrac{\text{height}}{\text{line of sight}},\ \cos\theta = \dfrac{\text{horizontal distance}}{\text{line of sight}},\ \tan\theta = \dfrac{\text{height}}{\text{horizontal distance}}.$

Standard values (angles in this chapter)

- $\sin 30^\circ = \frac{1}{2},\ \sin 45^\circ = \frac{1}{\sqrt{2}},\ \sin 60^\circ = \frac{\sqrt{3}}{2}.$
- $\cos 30^\circ = \frac{\sqrt{3}}{2},\ \cos 45^\circ = \frac{1}{\sqrt{2}},\ \cos 60^\circ = \frac{1}{2}.$
- $\tan 30^\circ = \frac{1}{\sqrt{3}},\ \tan 45^\circ = 1,\ \tan 60^\circ = \sqrt{3}.$
- $\sqrt{2} \approx 1.41,\ \sqrt{3} \approx 1.73$ (as given in the question).

Method

- Draw a neat figure; mark the right angle, the given angle and the unknown; choose the ratio containing the known and unknown sides.
- Two observation points in line with the tower: height h, distances $d_1 = \dfrac{h}{\tan\alpha},\ d_2 = \dfrac{h}{\tan\beta}$; distance between points $= |d_1 - d_2|$ (same side) or $d_1 + d_2$ (opposite sides).
- Add the height of the observer when the angle is measured from the eye.

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